Phylogenetic meta-analysis (metafor) – three views

1. Index form

The per-observation reading: what happens for each row i of your data.

$$
\begin{aligned}
y_i \mid \theta_i &\sim \text{Normal}(\theta_i,\, v_i), \quad v_i \text{ known} \\
\mu_i &= \beta_0 + u_{species(i)} \\
\mathbf{u}_{species} &\sim \mathcal{N}(\mathbf{0},\, \sigma^2_{species}\mathbf{A})
\end{aligned}
$$

2. Matrix form

The same model in matrix notation. The structural contract every textbook past chapter 4 switches to.

$$
\begin{aligned}
\mathbf{y} \mid \boldsymbol{\theta}, V &\sim \mathcal{N}(\boldsymbol{\theta},\, V), \quad V \text{ known} \\
\boldsymbol{\mu} &= \mathbf{X}\boldsymbol{\beta} + \mathbf{u} \\
\mathbf{u}_{species} &\sim \mathcal{N}(\mathbf{0},\, \sigma^2_{species}\mathbf{A}_{60\times 60})
\end{aligned}
$$

3. Worked observation (i = 1)

The index-form equations evaluated at the first observation of your data, with coefficient estimates plugged in.

$$
\underbrace{\text{yi}_1}_{\text{observed}=0.166} \quad = \; 0.374 \; - 0.201 \; - 0.00696 = 0.166
$$

Predicted $\hat{\mu}_1 = 0.173$*; residual* $\hat{\varepsilon}_1 = -0.00696$.